

AR Mining On Streams

Recap: Data streams

Stream of data arrive in rapid succession, no Re-scan is, storage space is insufficient to accommodate all data points.

Without storing all the data one wish to

- estimate Set of frequent items
- Number of distinct
- items Frequent
- itemsets

etc

[Constraints are on memory and processing power]

Frequent pattern mining over data streams

- Applications involves retail market data analysis, network monitoring, web usage mining, and stock market prediction.
- Using sliding window
- Efficiently remove the obsolete, old stream data
- Compact Pattern Stream tree (CPS-tree)
- 1
- Highly compact frequency-descending tree structure at runtime
- Efficient in terms of memory and time complexity
- Pane and window
- Insertion and restructuring

¹Tanbeer, Syed Khairuzzaman and Ahmed, Chowdhury Farhan and Jeong, Byeong-Soo and Lee, Young-Koo, "Sliding window-based frequent pattern mining over data streams", in Information sciences, 179(2,2) pages 3843–3865, Elsevier, 2009

CPS-tree construction

Algorithm 1 (Construction of a CPS-tree for a data stream)

Input: *Stream_data*, *Pane_size*, *Window_size*, *Initial_Sort_Order*

Output: T_{sort} : a CPS-tree for the current window

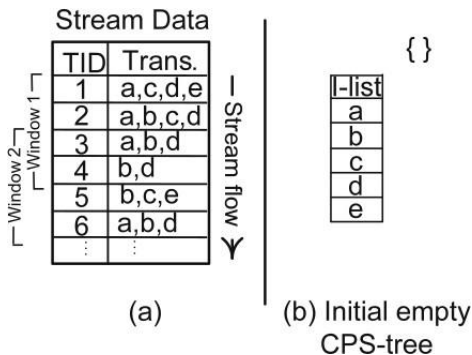
Method:

```
Begin
1:    $w \leftarrow \emptyset$ ;
2:    $T \leftarrow$  a prefix-tree with null initialization;
3:   Current_Sort_Order  $\leftarrow$  Initial_Sort_Order;
   //For the first Window
4:   While ( $w \neq$  Window_size) do
5:     Call Insert_Pane( $T$ ); // Insertion Phase
6:     Current_Sort_Order  $\leftarrow$  Frequency-descending sort order; // Restructuring Phase
7:     Restructure  $T$ ;
8:      $w = w + 1$ ;
9:   End While
   //At each slide of Window
10:  Repeat
11:    Delete the oldest pane information from  $T$ ; // Extracting the old pane
12:    Call Insert_Pane( $T$ ); // Insertion Phase
13:    Current_Sort_Order  $\leftarrow$  Frequency-descending sort order; // Restructuring Phase
14:    Restructure  $T$ ;
15:  End
End

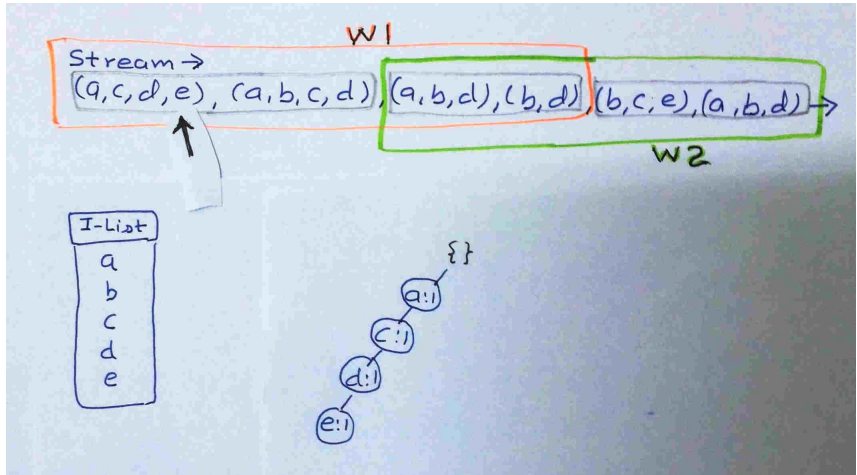
Insert_Pane( $Tr$ )
  Begin
1:     $p \leftarrow \emptyset$ ;
2:    While ( $p \neq$  Pane_size) do
3:      Scan transaction from the current location in Stream_data;
4:      Insert the scanned transaction into  $Tr$  according to Current_Sort_Order;
5:       $p = p + 1$ ;
6:    End While
  End
```

Fig. 3. The CPS-tree construction algorithm.

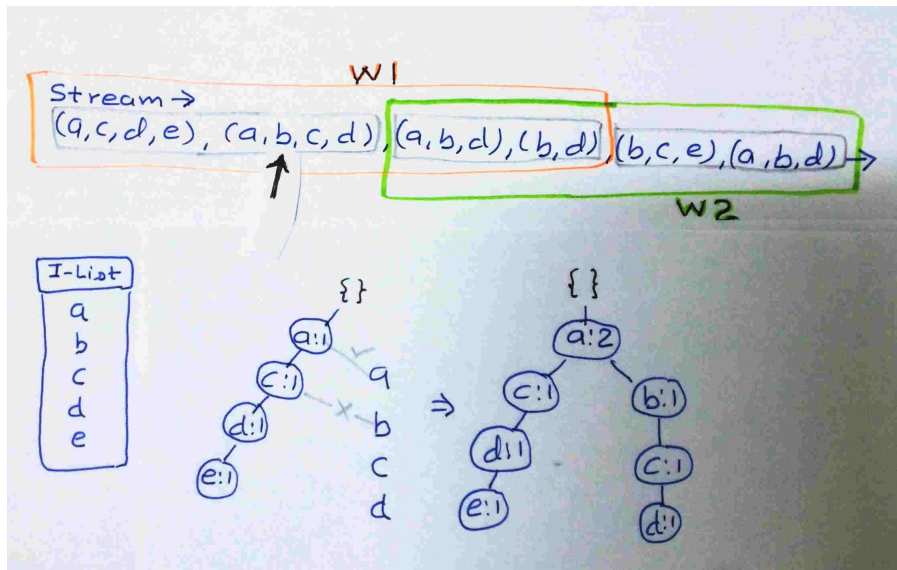
CPS-tree construction



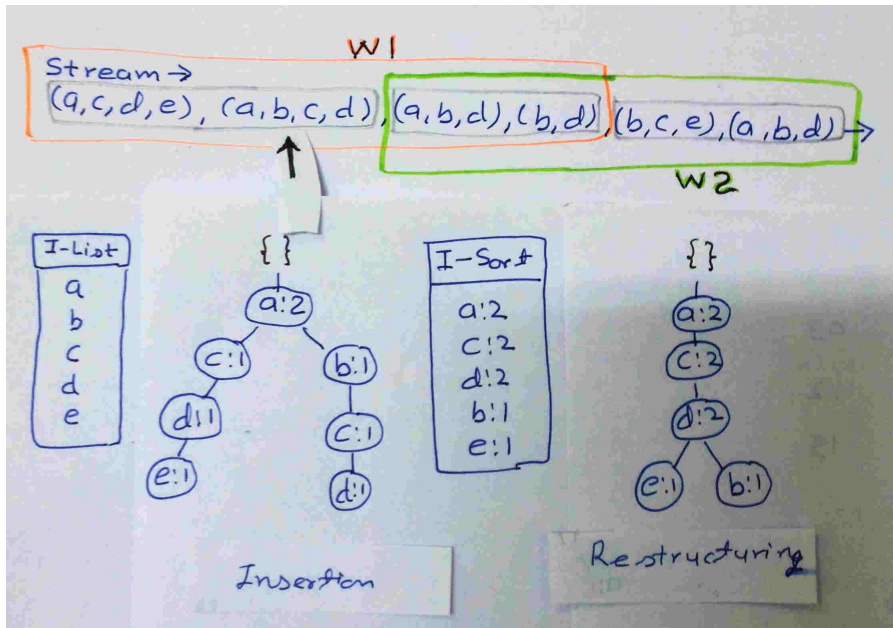
CPS-tree construction



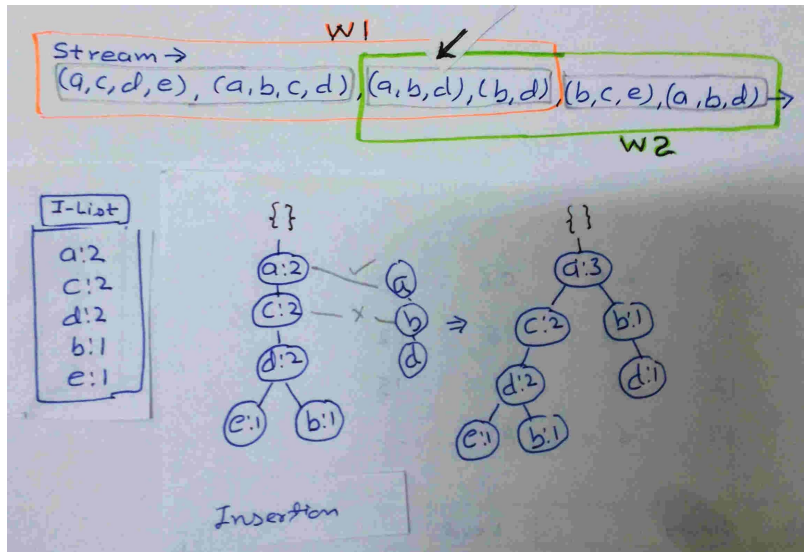
CPS-tree construction



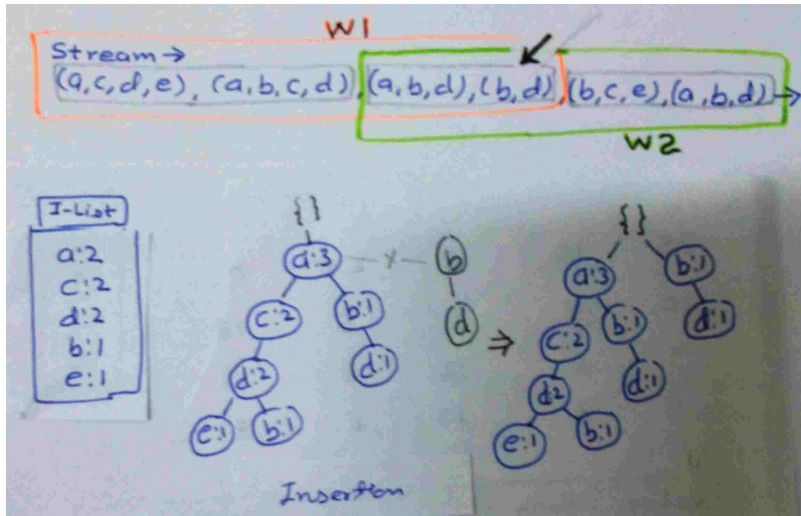
CPS-tree construction



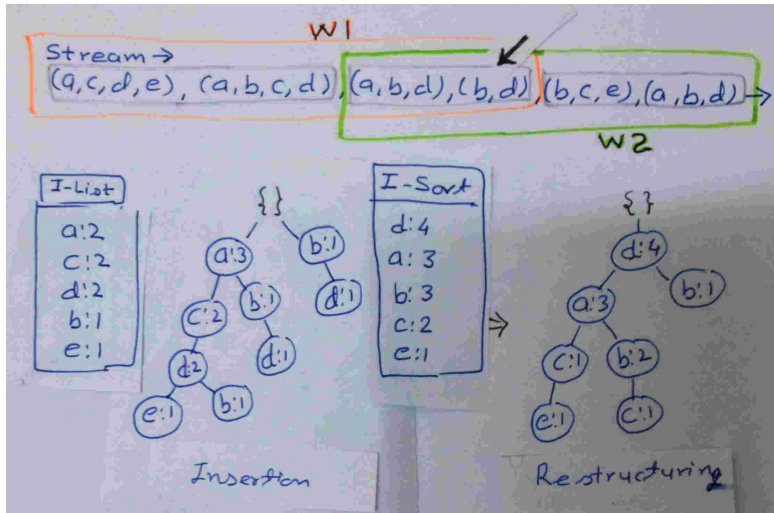
CPS-tree construction



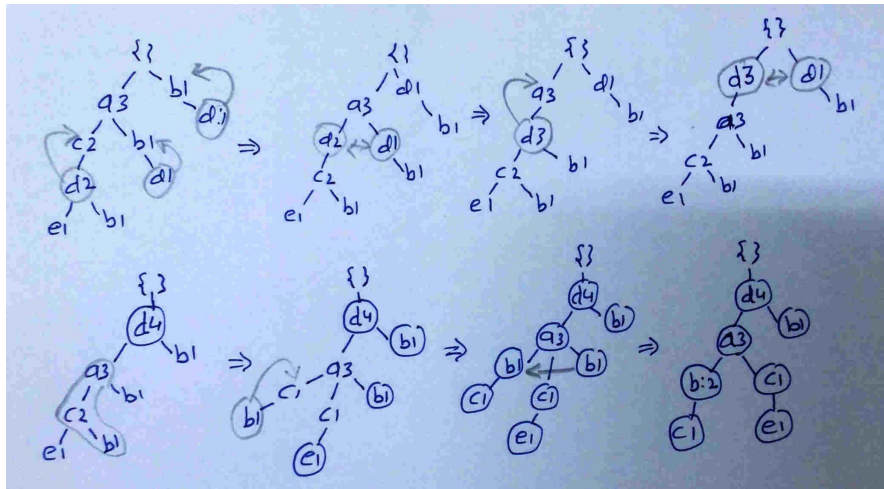
CPS-tree construction



CPS-tree construction

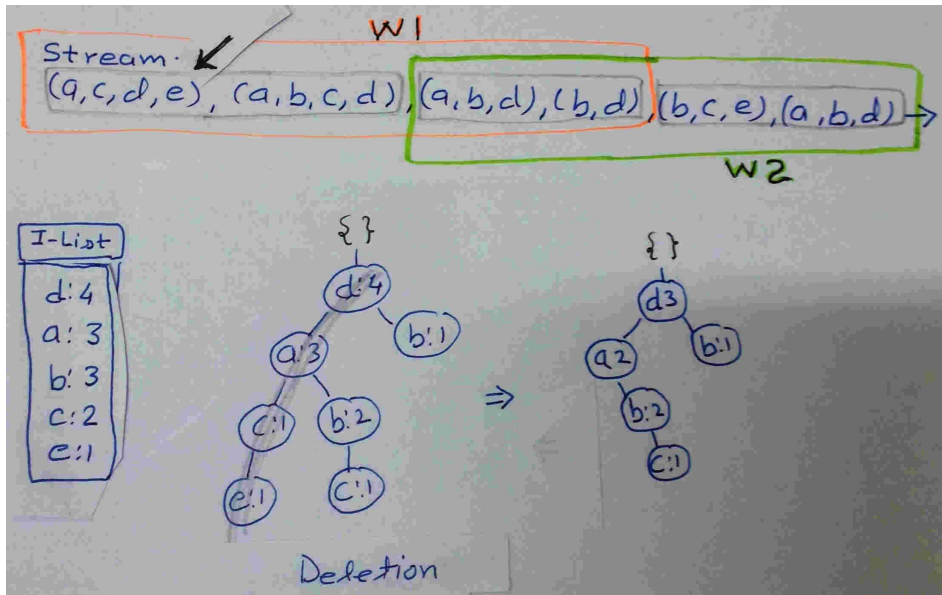


CPS-tree construction

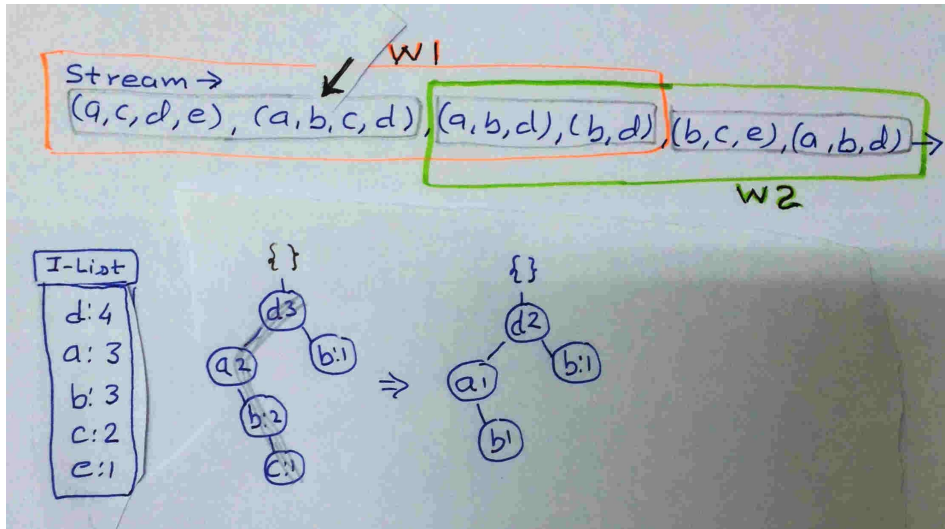


I-Sorted:
d,a,b,c,e

CPS-tree construction

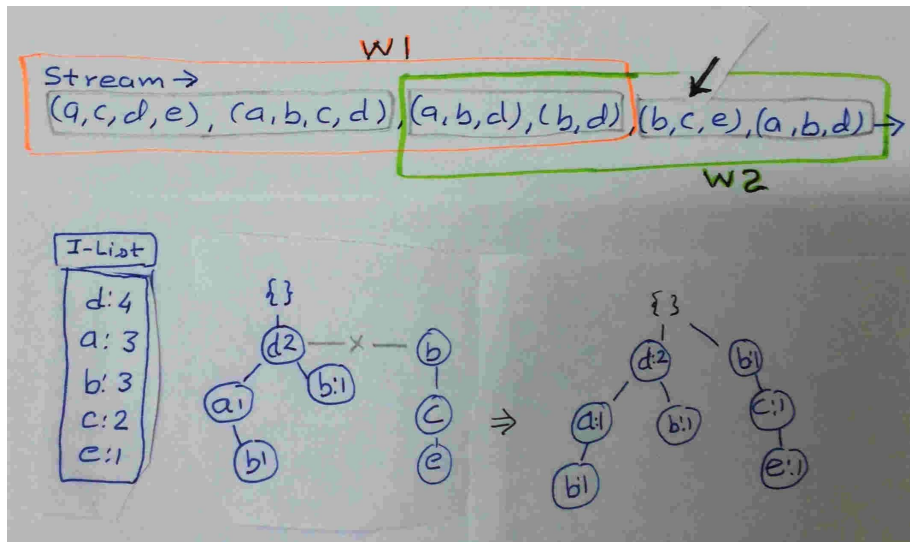


CPS-tree construction



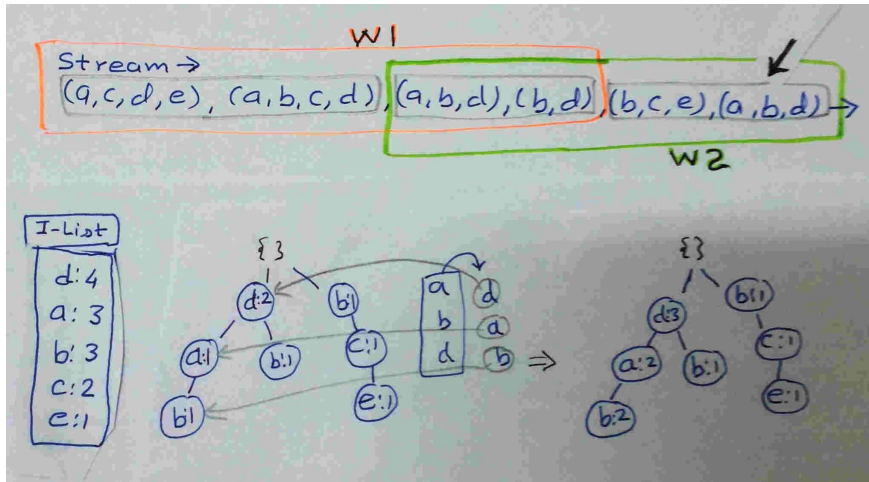
I-Sorted: d,a,b,c,e

CPS-tree construction



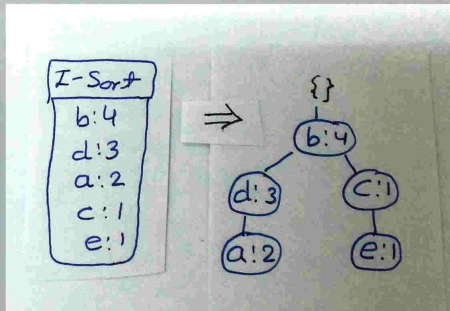
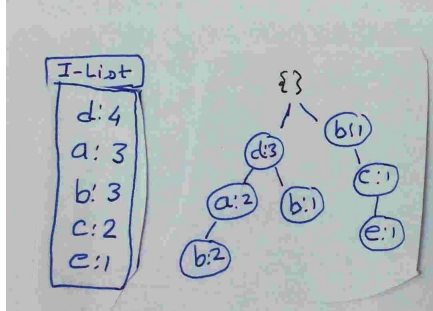
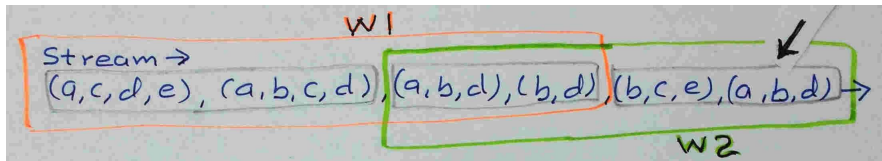
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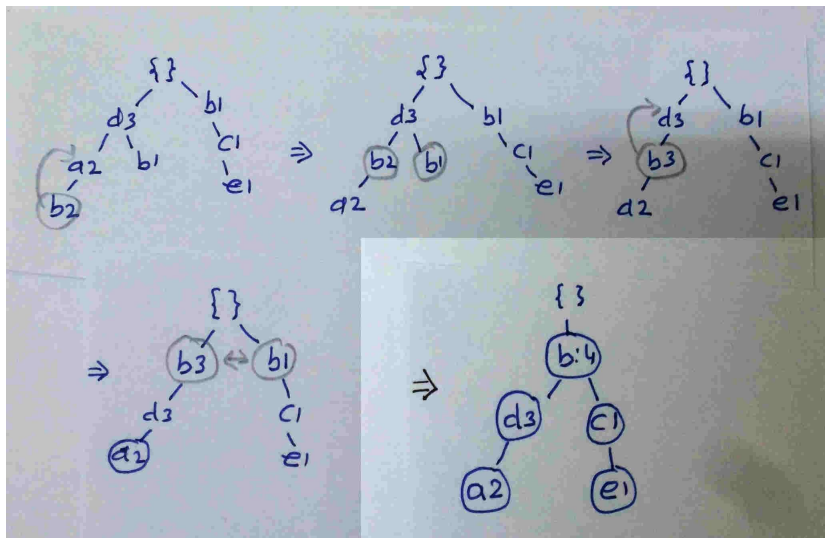
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d,a,b,c,e

CPS-tree construction



I-Sorted: b, d, a, c,
e

CPS-tree construction



I-Sorted: b, d, a, c,

CPS-tree Performance

